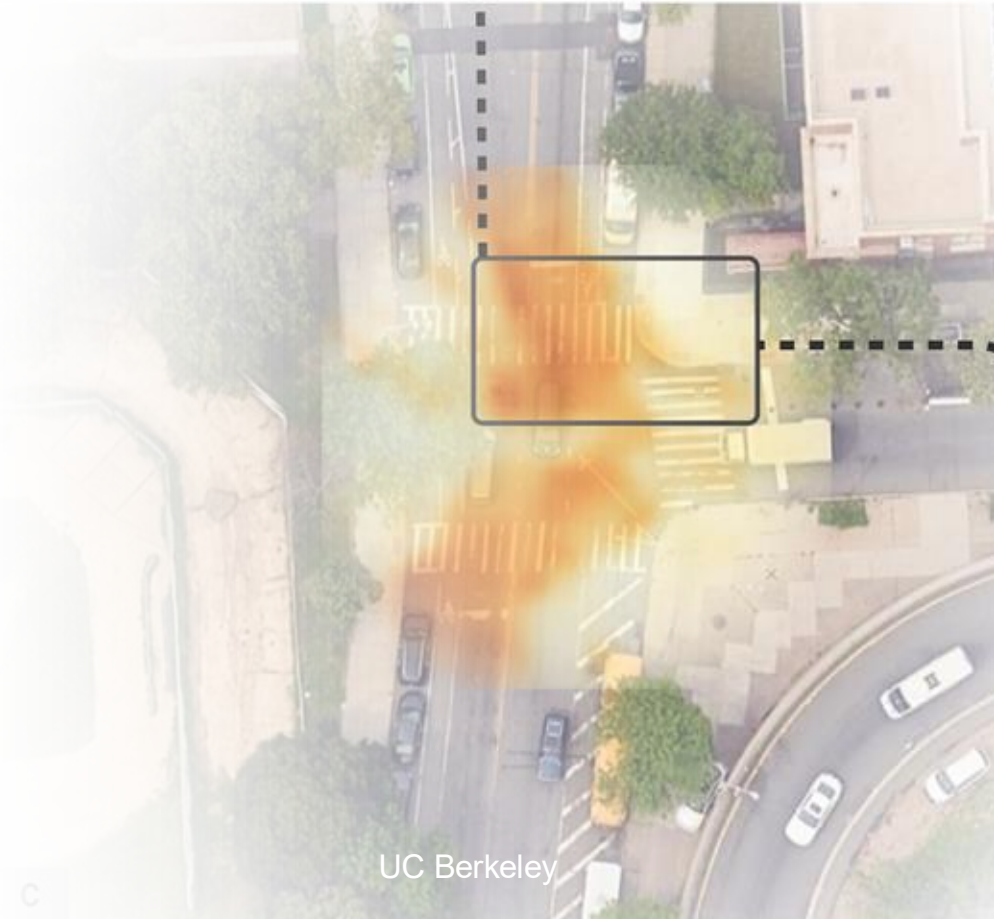
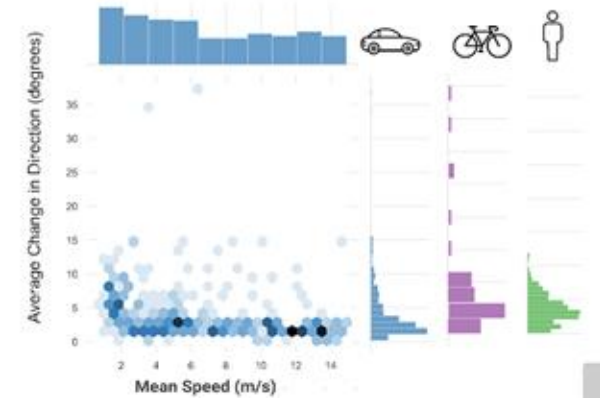
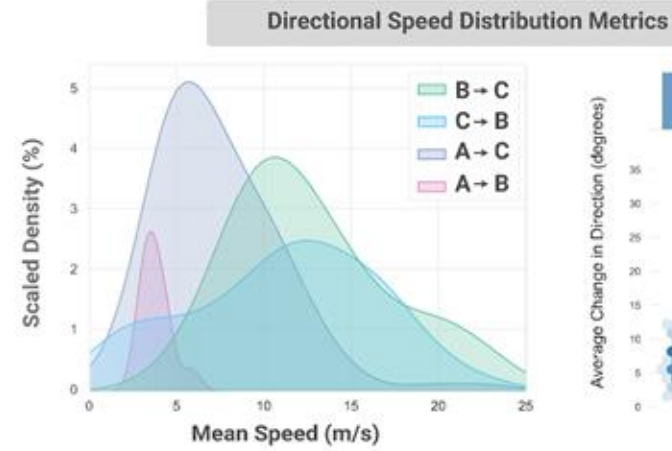


Introduction to Urban Data Analytics

Fall 2026

Maryam Hosseini

City & Regional Planning



UC Berkeley

A researcher who likes to code

- Background: urban systems, computer science, & economics
- MIT DUSP , NYU CS (VIDA lab)
- Open-source developer
- Research: Accessibility, computer vision, urban mobility, pedestrian safety, and heat exposure
- Designing interactive visual analytics tools



Berkeley
Boston
New York
Warsaw
Tehran







What is a city?

Cities are social institutions rather than physical artifacts.

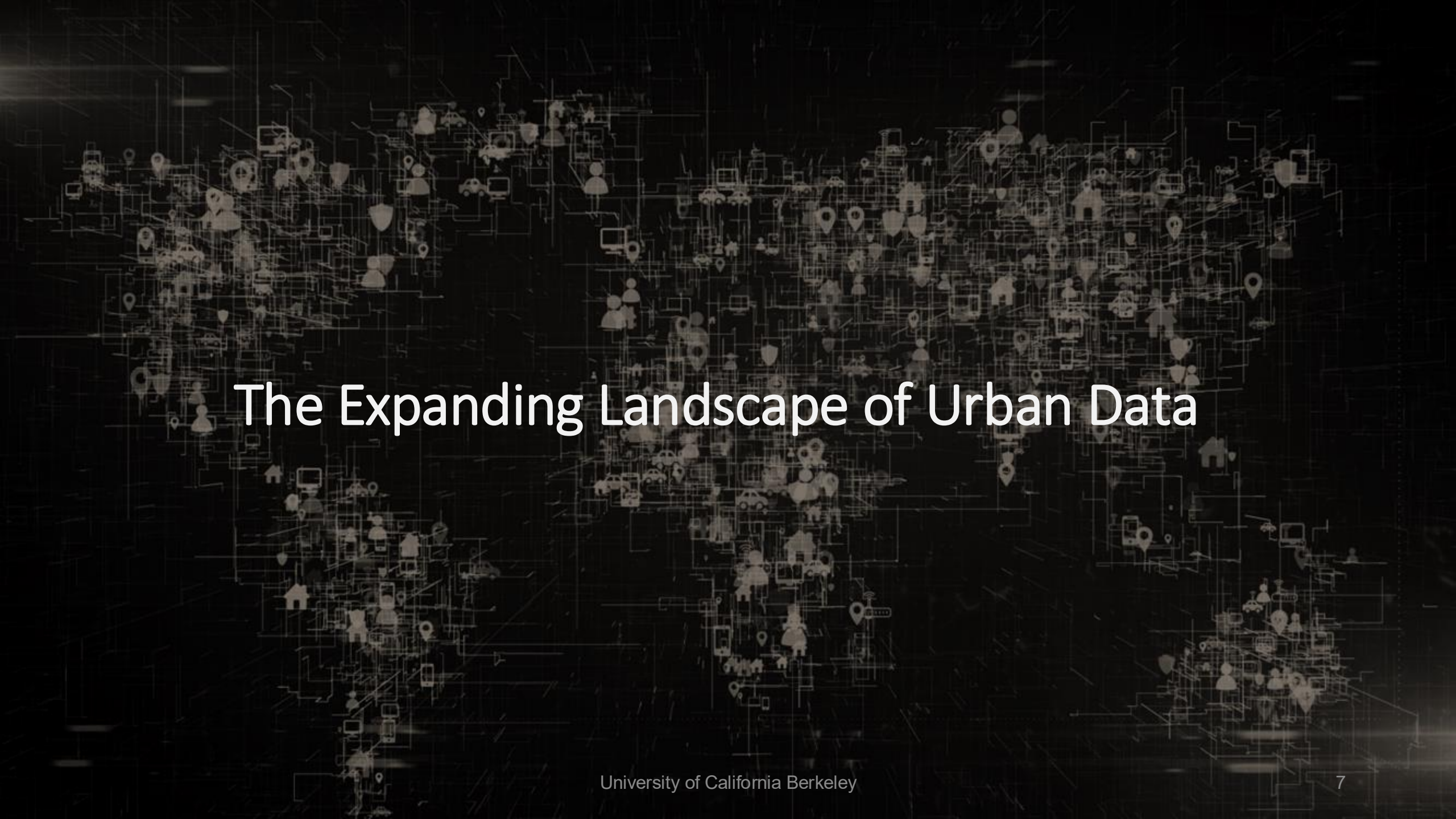
- Lewis Mumford

Cities are dynamic organisms, changing in response to how people interact with them.

—Jane Jacobs

For it is interaction, not place, that is the essence of the city and of city life.

- Melvin M. Webber



The Expanding Landscape of Urban Data

Where the sidewalk ends

Most cities don't map their own pedestrian networks. Now, researchers have built the first open-source tool to let planners do just that.

Peter Dizikes | MIT News Office
March 15, 2023



It's easier than ever to view maps of any place you'd like to go — by car, that is. By foot is another matter. Most cities and towns in the U.S. do not have sidewalk maps, and pedestrians are usually left to fend for themselves: Can you walk from your hotel to the restaurants on the other side of the highway? Is there a shortcut from downtown to the sports arena? And how do you get to that bus stop, anyway?

MIT researchers have built a source tool that uses aerial image-recognition to create sidewalks and crosswalks. Boston with an inset showing detected sidewalks, crosswalks. Image: Inset courtesy of the researchers via Google Earth



Mapping Sidewalks for Improved Connectivity

A new tool uses aerial image recognition to map a city's sidewalks and crosswalks. Its developers hope it will aid in creating a more comprehensive understanding of pedestrian networks and where improvements are needed.

1 Minute Read
March 22, 2023, 12:00 PM PDT
By Diana Ionescu @aworkoffiction

Deep learning model to classify crosswalks, sidewalks, and roads.

Deep learning package Living Atlas Subscriber

Item updated: Nov 3, 2025

View metadata

Owner:

Esri Analytics Team
esri_analytics

Details

Number of downloads
60,739

Size:
787.203 MB

Item ID

c0d520baa30d4b47ab362322...



Sharing

Everyone (public)

Acknowledgments

Hosseini, M., Sevtsuk, A., Miranda, F., Cesar, R. M., & Silva, C. T. (2023). Mapping the walk: A scalable computer vision approach for generating sidewalk network datasets from aerial imagery. Computers Environment and Urban Systems, 101, 101950.

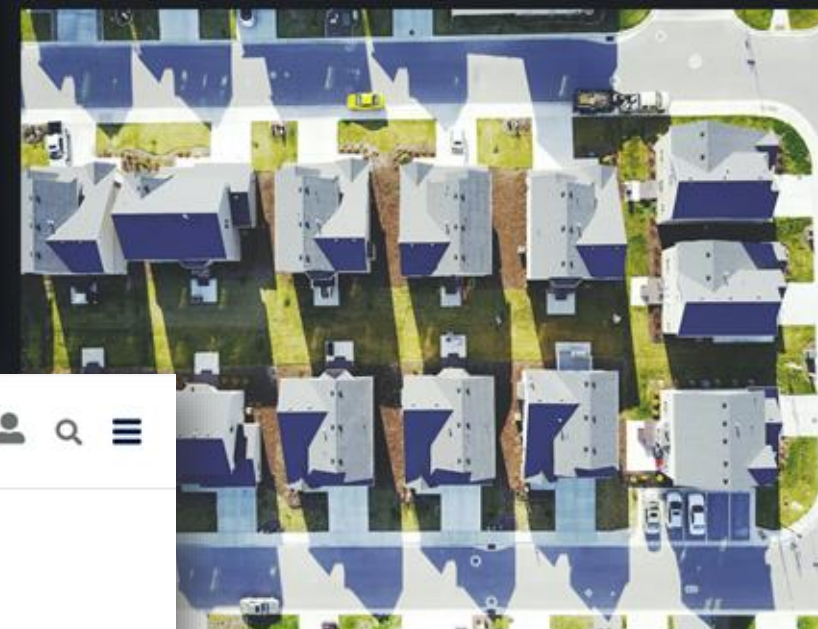
STREETSBLOG USA

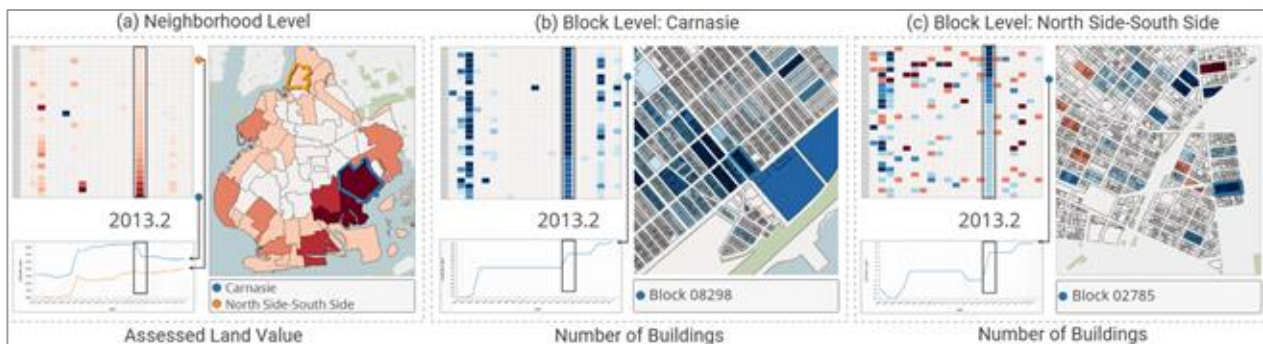
USA NYC MASS LA CHI SF CAL

MAPS

New Tool Maps Sidewalks From the Sky — And Encourages Cities to Fill The Gaps

6:42 AM EDT on March 27, 2023

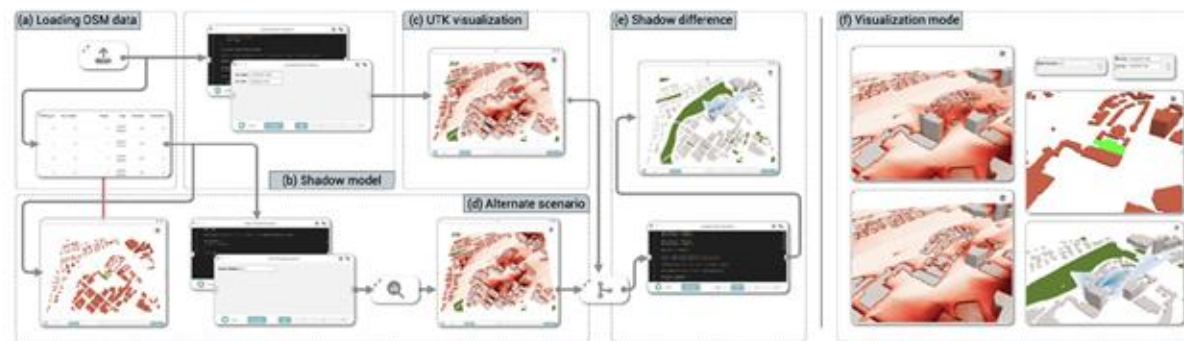




Urban Chronicles – Profiling City Land Use Evolution

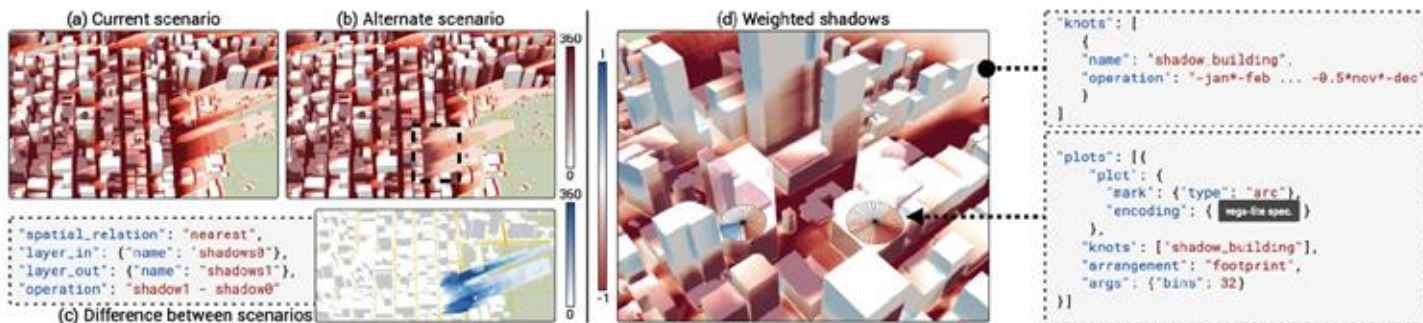
Curio: A Dataflow-Based Framework for Collaborative Urban Visual Analytics

Gustavo Moreira, Maryam Hosseini, Carolina Veiga, Lucas Alexandre, Nicola Colaninno, Daniel de Oliveira, Nivan Ferreira, Marcos Lage, and Fabio Miranda



The Urban Toolkit: A Grammar-based Framework for Urban Visual Analytics

Gustavo Moreira, Maryam Hosseini, Md Nafiul Alam Nipu, Marcos Lage, Nivan Ferreira, and Fabio Miranda



Urban Rhapsody

Why this class?

Why are You Taking this Class?

1. **Be intentional.** "Think" about why you are taking **each** course you've registered for.
2. **Set realistic goals.** Set 2-3 clear and attainable goals about what exactly you want to take out of each course you are taking and budget your time accordingly.

This Course Does Not Cover

- ArcGISPro / QGIS Software
- Statistical modeling, hypothesis testing
- Time series analysis
- Machine Learning
- Other Programming Languages (R, Java Script,)

This Course Does Cover

- Intro to urban Systems as a complex system
- Intro to GIS
- Intro to Computational Thinking
- Geospatial analysis in Python
- Data Visualization fundamentals
- Geospatial Data Visualization
- Deceptive Visualization & Misinformation

Checkout course schedule on the course website

Why learn programming?

A question of purpose

1. **Programming as an instrumental skill:** the emphasis is on the final product.

AI can do this

2. **Programming as a formative practice:** the emphasis is on developing reasoning, judgment, and the capacity to learn from failure.

AI cannot do it for us!

This **is not** an intro to programming course!

- Python basics (3 sessions)
- Transition to pandas and geopandas
- Principles of effective data visualization
- Creating expressive visualizations and maps in Python
- Exploring multiple libraries for geospatial visualization
- Hands-on analytics with real-world data
- Data sanity checks, cleaning, and implications
- Geospatial operations in Python

Weekly Time Commitment: **7.5 Hours** Outside Work

- **This course requires approximately **7.5 hours** of work outside of class each week.**
- Those who answered less than 3 hours: I recommend reconsidering your enrolment

Meet your GSI: Sean Flannelly

- 2nd Year **Master of City Planning**
- Worked for three years as a journalist in Greater Boston
- Interested in housing and transportation policy research
- Spent the summer interning with **the City of Oakland** and the **Turner Center for Housing Innovations**, working on qualitative research projects
- **Former CP255 student!**



Meet your GSI: Evan Kartheiser

- 2nd Year **Master of Information and Systems** at the School of Information
- Interested in applying remote sensing to climate and environmental problems
- CITRIS Responsible Technology Fellow at **CalEPA** where I supported the agency's open data portal
- Former **D-Lab** consultant advising Berkeley researchers on geospatial data analysis projects



Course Logistics



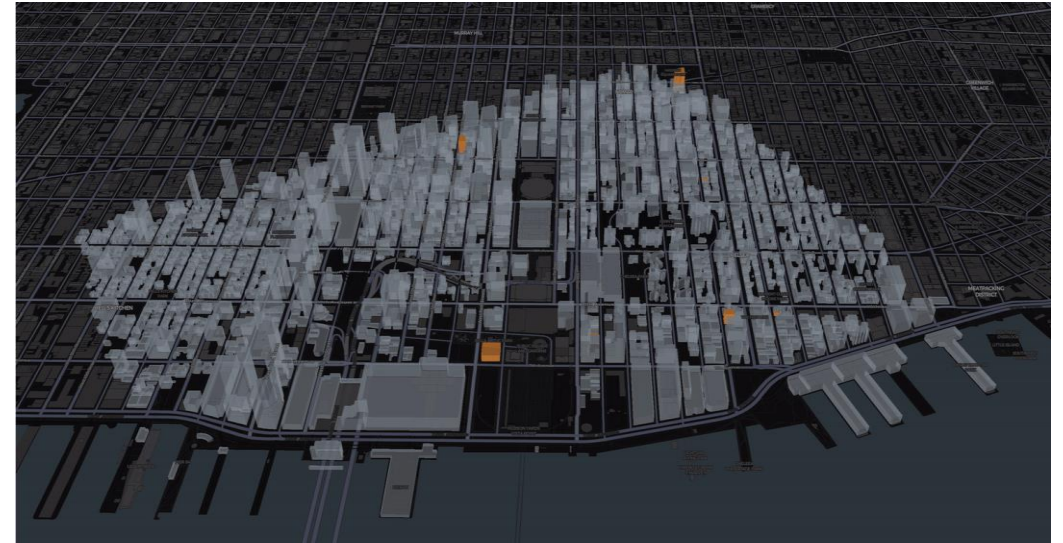
Attendance, Assignments, Final Project

Course Structure: Three Modules

1. Foundations

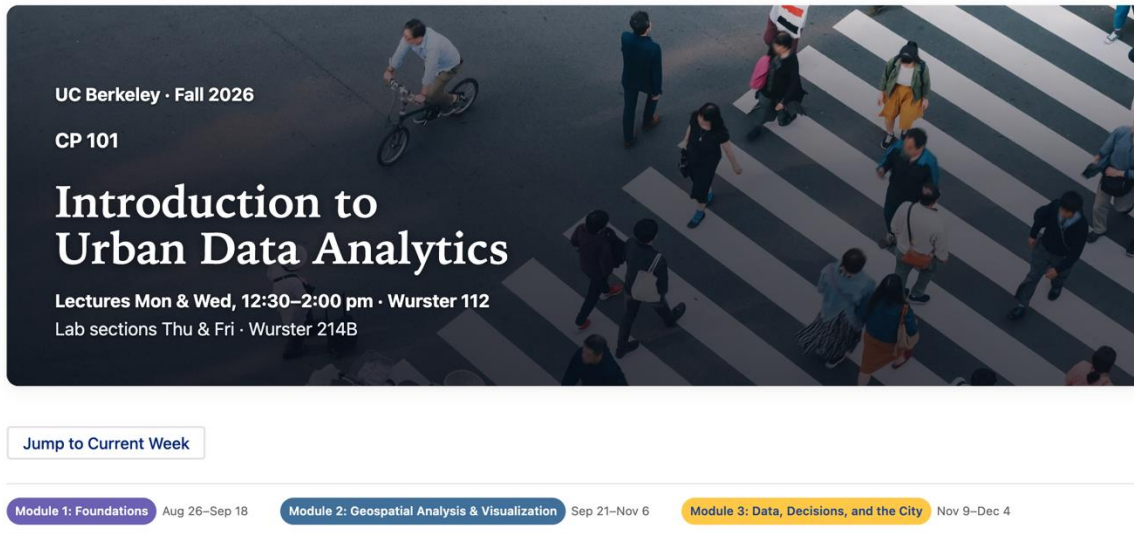
2. Spatial Analysis & Visualization

3. Data, Decisions, and the City



Course Website

<https://ucb-cp101.github.io/fa26/>



Main source of information about this course. The site will be updated frequently.

- **Bookmark It:** Add the website to your bookmarks and check it regularly for announcements and materials.
- **Readings and Resources:** Lecture slides, reading and videos and extra materials
- **Assignments, Labs and Class Activities**
- **Course Policies and Syllabus**

Grading Breakdown

- **Class attendance and activities (15%)**
- **Lab attendance and activities (15%)**
- **Assignments (4 Graded): 30%**
 - Dropping the lowest submission
 - **Unsubmitted work/empty submission receives zero and cannot be dropped.**
- **Final Project: 40%**
 - Proposal : 5%
 - Midterm Submission: 15%
 - Final Submission: 20%

Late Submission Policy

Extensions

- 2 extension tokens (24hrs each) for **assignments**
- 1 extension token (72hrs) **final project**

Late Submissions (Without Extensions)

- **Penalty: 10%** deduction **per day** will be applied to late submissions without an approved extension.

Attendance Policy

- Your presence is important!
- You get up to **4 absence** no questions asked
- Class presence and activities
have **15%** of your **final grade!**

Final Project

Track A: Python for Data Interpretation & Visual Reasoning

For students interested in interpreting urban data through maps and visual analysis. Lighter coding focus.

Deliverables

- ArcGIS StoryMap (or equivalent)
- 4+ standalone interactive HTML maps generated with Python
- A PDF report

Emphasis

- Visual reasoning and interpretability
- Thoughtful use of spatial methods, robust cleaning process
- Problem definition, story telling and visualization design judgment

Track B: Python for Data Analytics & Reproducibility

For students more comfortable coding in Python and aiming to showcase technical skills. Shorter written reports in interim phases.

Deliverables

- GitHub repository + GitHub Pages webpage
- 2+ standalone interactive and animated HTML maps generated with Python (integrated in the webpage)

Emphasis

- Reproducible, modular workflows
- Handling large or more complex datasets
- Robust analytical methods

Opportunity Strong projects may be developed into workshop submissions, conference posters, or journal publications.

Final Project Due Dates

1. Group Formation

Sep 9 – 12:00 pm

2. Proposal and Lightning Talk Video

Sep 21 – 12:00 pm

3. Midterm Submission

Nov 2 – 11:59 pm

4. Final Project Submission

Dec 15 – 11:59 pm

GenAI Policy

You can use any GenAI tools you like but use them wisely!

- Request references in your prompt
- Verify responses and refs with trusted sources
- **You must understand** the code you submit
- **Type the code instead of copying and pasting**

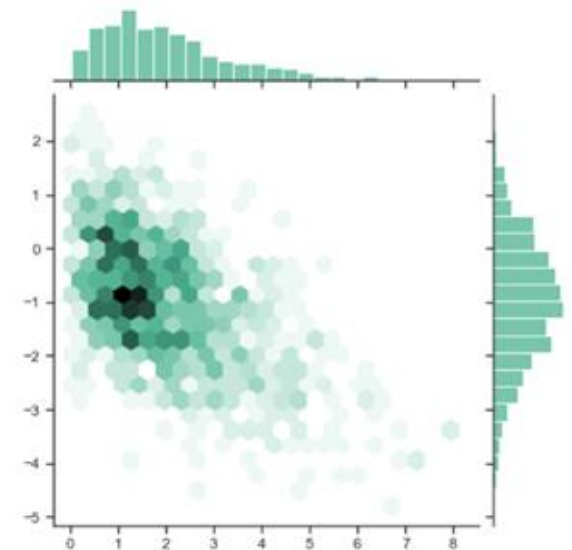
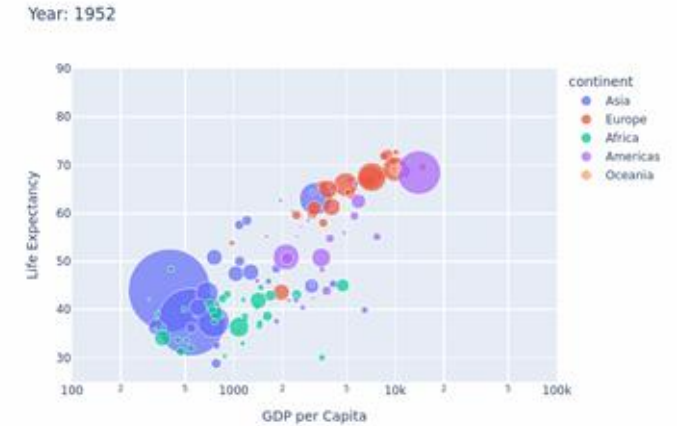
Don't let any tool take away your joy of problem-solving!

Guest Speakers

- **Lucy Andrews, Senior Environmental Scientist, CA Department of Water Resources**
Data Ethics and Governance | Oct 12
- **Sam Pottinger, Senior Research Data Scientist / Software Engineer, DSE**
Visualization, Interactivity, and Co-design | Oct 14
- **Jack Moorehead, DCRP**
Land Surveys and Mapping the City | Oct 26
- **Min Ragan-Kelley, Founding member of the Jupyter Steering Council & Co-creator of JupyterHub, BIDS**
Interactive Visual Analytics Workshop | Nov 2
- **Anwar Baroudi, Oakland Mayor's Commission on Persons with Disabilities (MCPD) Chair, DCRP**
Accessibility, Equity, and the Built Environment | Nov 4
- **Sasha Blair-Goldensohn, Staff Accessibility Lead, Google Maps**
17 Years on Wheels: Subways, Maps, Handcycles, and Disability Pride | Nov 9

Infinite Possibilities

- **Powerful Visualization Tools:**
Work with tools like **Plotly** and **Lonboard** to create, fast, interactive visualizations
- **Critical Thinking in Visualization:**
Learn about **deceptive visualizations & misinformation** and develop skills to spot them quickly and effectively.



Practice is the key to success in this course.

MTC Collaboration